I · Monitor mount and accessories

The VIVO STAND-V300CU (P12) has two independent C-clamps and no base bar. Both clamp onto top A's own wall edges, never on the joint: clamp 1 centred at **x = 40.0 cm** on the back edge, clamp 2 centred at **y = 40.0 cm** on the side-wall end. The crossbar runs on the diagonal between the poles. Each lower jaw bears on a spare steel mending plate (P11), because particleboard crushes under a bare steel jaw over months. Datum: top A's back-left corner (0, 0); x along the back wall, y along the side wall.

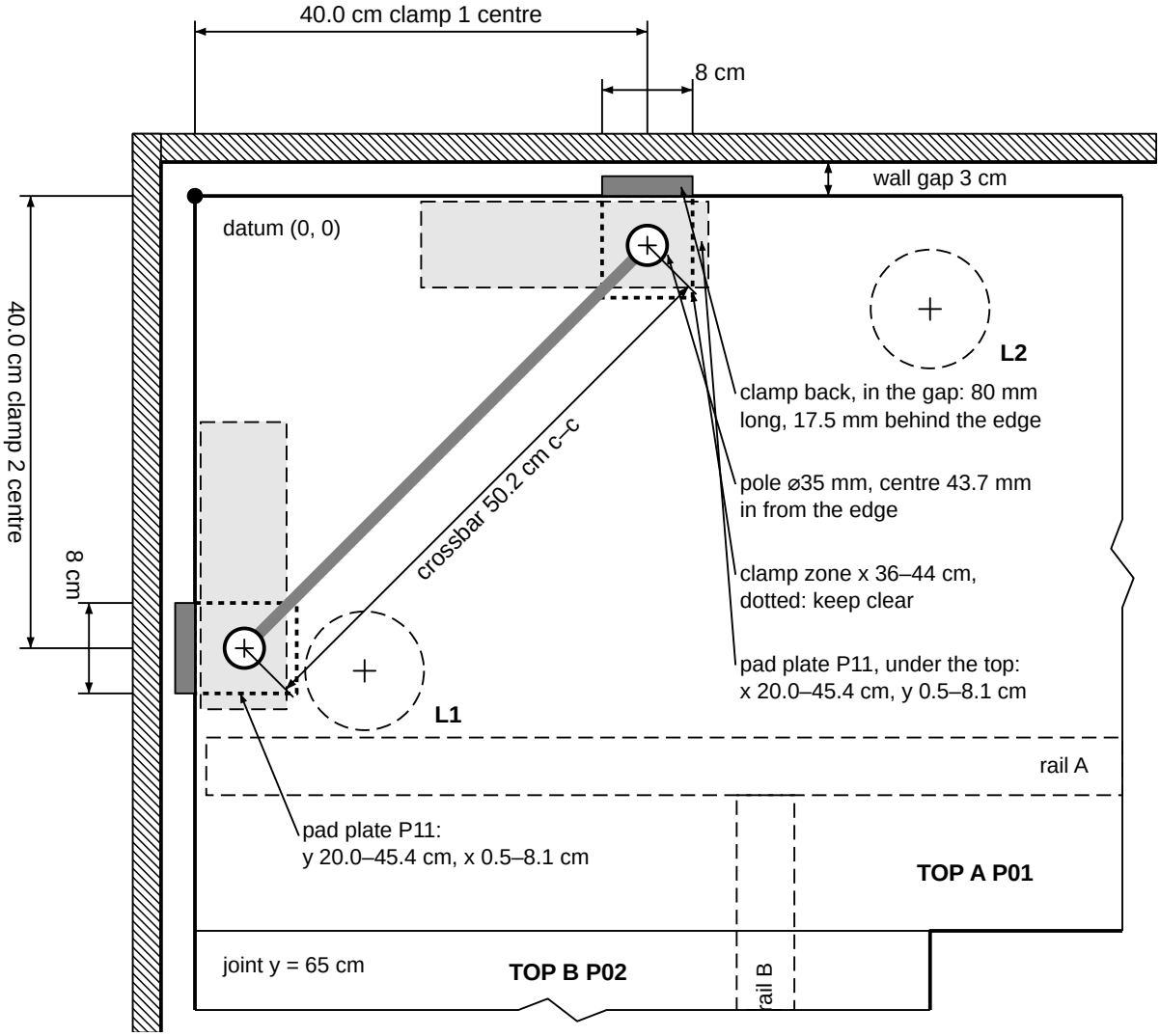


Fig. I-1 Plan of the corner from above; tape readings in cm, caliper sizes marked mm. Dashed: under the top (pad plates P11, legs L1 and L2, rails). Dotted: the clamp zones, 36–44 cm along the edge, keep clear. Dark: the clamp backs in the 3 cm wall gap. Black circles: poles, centre 43.7 mm in from the edge.

Balloon	Part	Balloon	Part
P01	Top A, EKBACKEN 249 × 65 cm, 28 mm thick	P11	Mending plate 10 × 3 × 0.12 in, two used here as clamp pads
P02	Top B, EKBACKEN 249 × 65 cm, 28 mm thick	P12	VIVO STAND-V300CU (clamps, poles, crossbar, arms)

Contents

I · Monitor mount and accessories 1

K · Final inspection and load testing 7

L · Maintenance and troubleshooting 13

Appendix · Outstanding measurements 15

What is measured at the clamp

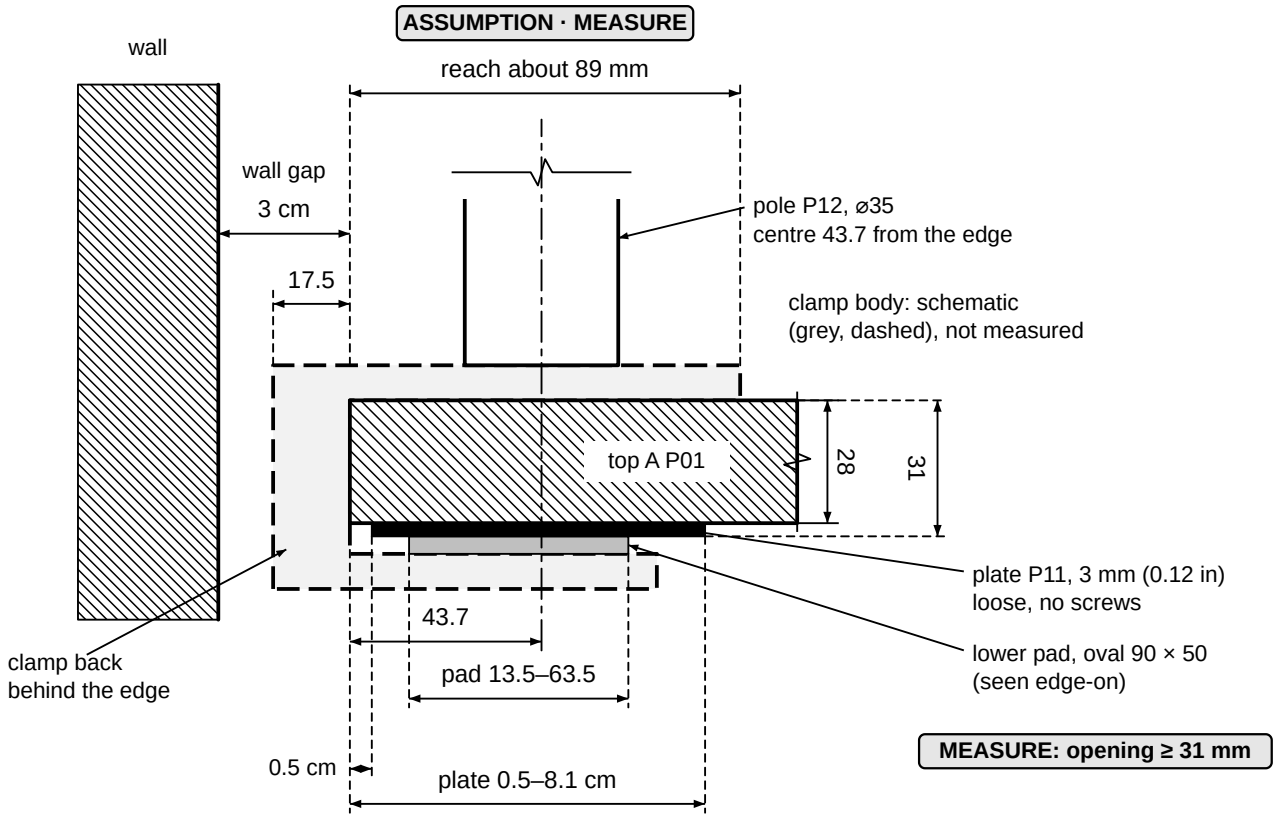


Fig. I-2 Section through clamp 1 at x = 40.0 cm, looking along the back wall; mm unless marked cm. The clamp body is drawn as a plain C (grey, dashed) because its shape is not measured; only the dimensioned values are. The section through clamp 2 at y = 40.0 cm is the same, turned 90°.

Value	Size	Confidence
Crossbar, centre to centre of its end collars (19¾ in)	50.2 cm	VERIFIED
Pole diameter; pole centre in from the desk edge	35 mm; 43.7 mm	VERIFIED
Clamp back: length along the edge; projection behind the edge (it sits in the 3 cm wall gap)	80 mm; 17.5 mm	VERIFIED
Lower pad: oval 90 × 50 mm, 13.5 to 63.5 mm in from the edge, so it lands on the plate (0.5–8.1 cm from the edge)	13.5–63.5 mm	VERIFIED
Clamp centre from the corner: ideal 39.9 cm; built at 40.0 cm, the crossbar sets the last millimetre	40.0 cm	SOURCE-DEFINED
Clamp zones x 36–44 cm, y 0–9 cm (clamp 1) and y 36–44 cm, x 0–9 cm (clamp 2); pad plates x 20.0–45.4 cm, y 0.5–8.1 cm and y 20.0–45.4 cm, x 0.5–8.1 cm	Fig. I-1	SOURCE-DEFINED
Top 28 mm plus plate 3 mm (0.12 in): under 60 mm, so VIVO's Option 1 holes (2 and 4 on the lower clamp)	31 mm	SOURCE-DEFINED
Top arm reach from the edge	about 89 mm	ASSUMPTION / FIELD CHECK
Maximum jaw opening (must be 31 mm or more)	≥ 31 mm	MEASURE ON HARDWARE

Hang the mount

1 **Lay the pad plates.** Lay a spare mending plate P11 flat on the underside, on each pad outline drawn when top A was marked (chapter D), long side along the edge: x 20.0–45.4 cm, y 0.5–8.1 cm at clamp 1; y 20.0–45.4 cm, x 0.5–8.1 cm at clamp 2. Loose, no screws: the jaw holds it. SOURCE-DEFINED

> ! **MEASURE ON HARDWARE before hanging the clamps**
> Top arm reach (about 89 mm) and maximum opening (needs 31 mm or more: 28 mm top + 3 mm plate) are not measured. Open the lower clamp fully on the Option 1 holes and measure the gap with calipers; if it is under 31 mm, stop. The clamp zones are drawn 9 cm deep from the edge; if the reach measures more, keep the zone clear to the measured reach.

2 **Hang both lower clamps, finger-tight only.** Clamp 1 centred at x = 40.0 cm on A's back edge, clamp 2 centred at y = 40.0 cm on A's side-wall end (Fig. I-1). Use the VIVO sheet's Option 1 holes (2 and 4): top and plate together are 31 mm, under 60 mm. The clamp backs sit in the 3 cm wall gap; the lower pad lands on the plate, 13.5–63.5 mm from the edge (Fig. I-2).

3 **Fit the poles and the crossbar as the spacing guide** (VIVO sheet step 3). The crossbar measured 50.2 cm centre to centre, so it fixes the pole spacing. **Only now tighten both clamps**, with the crossbar on. VERIFIED

4 **Refit the crossbar at monitor height** (VIVO sheet step 6): 40–43 cm above the top (123.0–126.0 cm from the floor).

5 **Hang the three screens, all landscape**, and set the heights in Fig. I-3. The G5 goes on the centre bracket (VESA 75), screen centred on the diagonal at (36.5, 36.5) cm, 20 cm in front of the crossbar. The Acers go on the side arms, toed in 15°; each Acer's near end ends up about 12 cm from the G5's curved end. The side arms run along the wall behind each panel, then forward into the VESA plate.

Weights against the mount's ratings

Screen	On the mount	Rating	Confidence
Samsung Odyssey G5 32 in P13 (1000R, VESA 75), centre	about 5 kg	centre bracket 48.4 lb curved (59.4 lb flat); top edge at most 26.5 in (67.3 cm) above the top	SOURCE-DEFINED
Acer XZ320QU 31.5 in P14 (1500R, VESA 100), each side	about 5 kg (5.6 kg with its stand)	17.6 lb (8 kg) per side arm	SOURCE-DEFINED

Ratings: VIVO's listing as quoted in docs/design-review.md (triple-monitor mount); screen weights from the same review.

> The clamps load top A's edges near legs L1 and L2, not the corner tip. Do not add a soft pad under a jaw: the steel plate is the pad.

Heights above the top

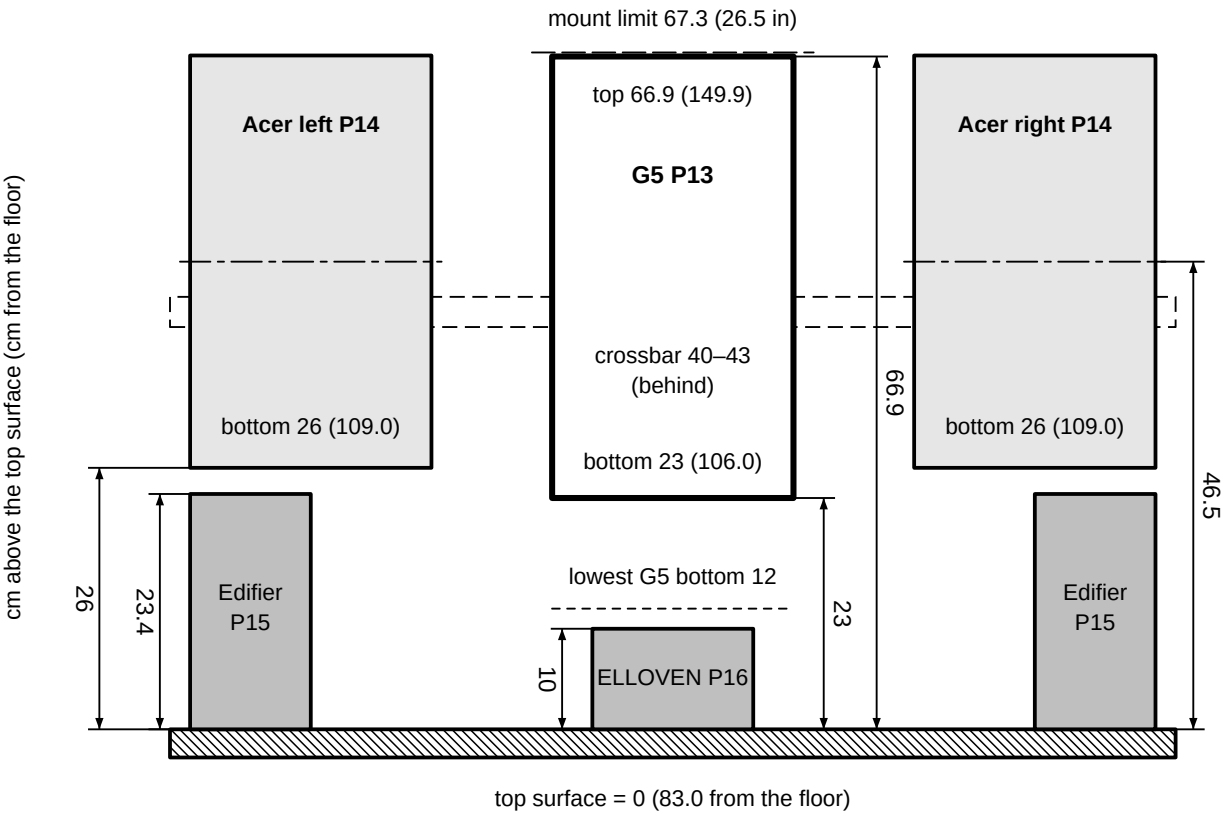


Fig. I-3 The three screens seen from the seat, unfolded flat. Heights to scale, in cm above the top surface (in brackets, cm from the floor); widths are schematic. The Edifiers stand under the outer half of each Acer.

Item (all cm)	Above the top	From the floor	Note
Top surface	0	83.0	underside 80.2, top 28 mm thick
ELLOVEN P16 top	10	93.0	13 under the G5
G5 bottom edge	23	106.0	SOURCE-DEFINED
Edifier P15 top	23.4	106.4	2.6 under each Acer
Acer bottom edges	26	109.0	panel centre 46.5 above the top
VIVO crossbar	40–43	123.0–126.0	refitted at monitor height
G5 top edge	66.9	149.9	0.4 under the mount's 67.3 (26.5 in) limit

IF THE ASSEMBLED MOUNT IS LOWER
The G5's top edge at 66.9 cm is only 0.4 cm under the 67.3 cm limit. If the assembled mount will not reach, lower the G5: a bottom edge anywhere down to 12 cm above the top still clears the ELLOVEN (10 cm tall).

Everything else on the desk

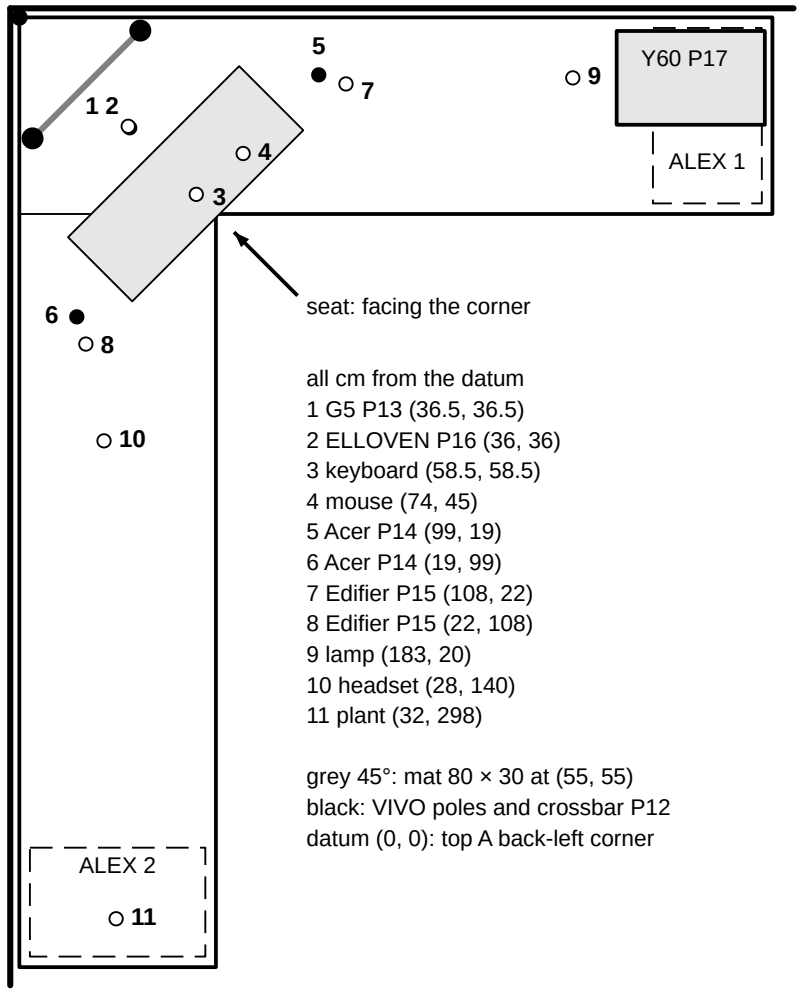


Fig. I-4 Plan of the desk, centres in cm from the datum, top A's back-left corner. Dashed: ALEX 1 and ALEX 2.

ELLOVEN at (36, 36) cm, turned 45°, drawer toward you: its back corners clear the clamp plates and poles by over 2 cm. At (33, 33) cm it would sit on the clamps.

The mat's back edge touches the ELLOVEN's front feet; its front edge reaches the inside corner of the L.

The Y60 (P17) stands on its I/O end on the printed pedestal over ALEX 1: long axis along the back wall, side glass toward you, front glass up. Keep it there, not mid-wing.

Item	Centre (x, y), cm	How
G5 P13	(36.5, 36.5)	on the diagonal, 20 cm in front of the crossbar
ELLOVEN P16	(36, 36)	turned 45°, drawer toward you
Deskmat 80 × 30 cm	(55, 55)	turned 45°
NuPhy Air75 V2; mouse	(58.5, 58.5); (74, 45)	keyboard turned 45°, on the mat
Acer P14, right; left	(99, 19); (19, 99)	along each wall, toed 15° toward you
Edifier P15 ×2	(108, 22); (22, 108)	grilles toward you, toed with the Acers
HYTE Y60 P17	x 197.5–246.5, y 4.5–35.5	on its pedestal, over ALEX 1
ELYONA lamp	(183, 20)	beside the Y60, arm toward the keyboard
Headset stand; plant	(28, 140); (32, 298)	on top B, hook toward you; plant over ALEX 2

All positions **SOURCE-DEFINED** (docs/build-manual.md step 8).

K · Final inspection and load testing

Ten checks, in order, before you call the desk done. Each is an ACCEPTANCE CHECK: a pass or fail you can see or measure. Several of them test a STRUCTURAL ESTIMATE from the design review that has not been measured on this desk; the check says which. Write the results in the boxes and in the sign-off table at the end of this chapter. Datum for every drawing: top A's back-left corner (0, 0).

Check 1 · Surface height at the six corners ACCEPTANCE CHECK

1

Measure from the floor to the top face at the six supported corners 1 to 6, then the inside corner 7 last (Fig. K-1). Every reading **83.0 cm ± 0.3 cm** (82.7 to 83.3 cm). With the optional ALEX feet fitted, read instead: 2 cm feet 85.0 cm, 4 cm feet 87.0 cm, 6 cm feet 89.0 cm, each ± 0.3 cm. If a corner is out, adjust only what carries it, then measure all seven again (positions x, y in cm): corner 1 (0, 0) by legs L1 and L2; corners 2 (249, 0) and 3 (249, 65) by the ALEX 1 feet; corners 4 (65, 314) and 5 (0, 314) by the ALEX 2 feet; corner 6 (0, 65) by leg L4. Corner 7, the inside corner (65, 65), has no support of its own: it follows the joint, so it is measured last and set by chapter H, never adjusted here.

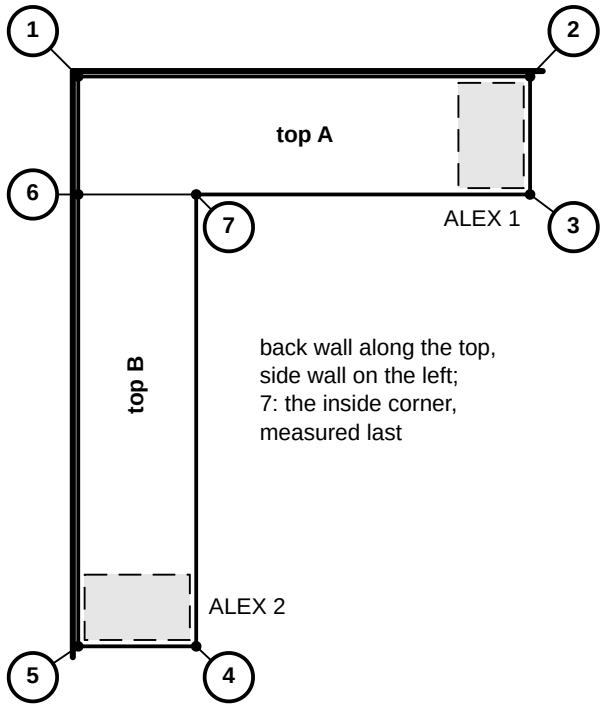


Fig. K-1a Corners 1–6, and the inside corner 7 measured last.

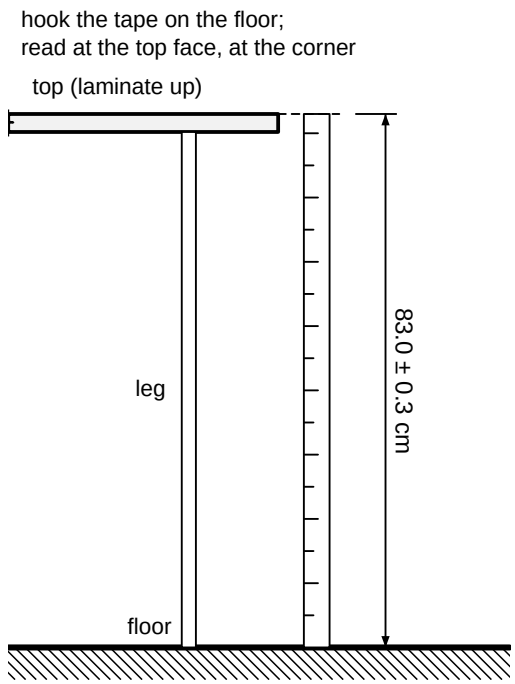


Fig. K-1b Tape from the floor to the top face, at the corner.

Corner	1	2	3	4	5	6	7 (last)
Reading (cm)							

Check 2 · Level and flush across the joint ACCEPTANCE CHECK

2

A level across the joint reads flat both ways: across it (1) and along it (2). A fingertip drawn across the joint (3) finds no step. Level the two tops with the OLOV feet and the round feet, never by packing anything under the joining plates.

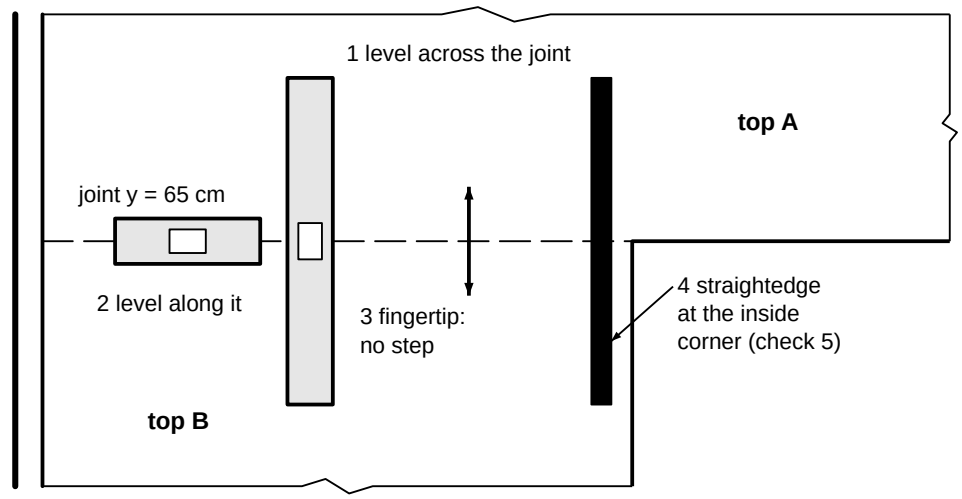


Fig. K-2 Plan of the joint at y = 65 cm. 1, 2: level positions; 3: fingertip; 4: where the straightedge of check 5 goes, at the inside corner.

For the remaining checks you need



Tape
floor to top face, checks 1 and 8



Level, straightedge
checks 2, 3 and 5; a rule to read the gap



20 kg
a case of water plus a few books, check 3



Wrench
torque wrench at 25 N·m, or a 15 cm wrench, checks 5 and 10

Do the checks in this order, with the screens and everything else in place.

Check 3 · Corner load test ACCEPTANCE CHECK

> **NEVER STAND OR SIT ON THE DESK**
> The test load is a known weight: 20 kg of water and books. Do not stand or sit on the corner; that is not what the beam was sized for.

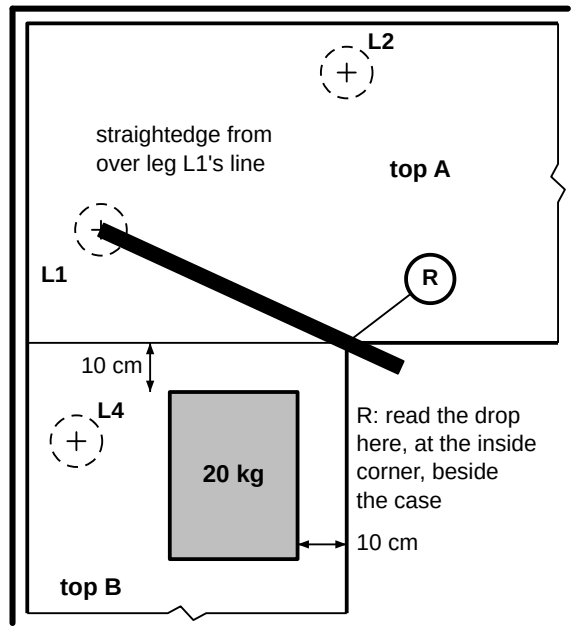


Fig. K-3a Plan. Case faces 10 cm in from top B's front edge and from the joint (case size schematic). R: reading point.

3

Set **20 kg** (a case of water plus a few books) on top B's inside corner, **10 cm in from both edges**.

Rest a straightedge from leg L1's line across the corner. Read the drop at the corner (R): **no more than 2 mm**. The estimate is about 1 mm. STRUCTURAL ESTIMATE

Lift the weight off. The corner returns to where it was, nothing creaks, no screw or bolt has moved.

Tests: sag at the inside corner (rail A carrying rail B's corner reaction) and corner-joint slip (the six fitting bolts holding by friction). STRUCTURAL ESTIMATE

Drop under 20 kg (mm)	Returned?	Creak / movement?

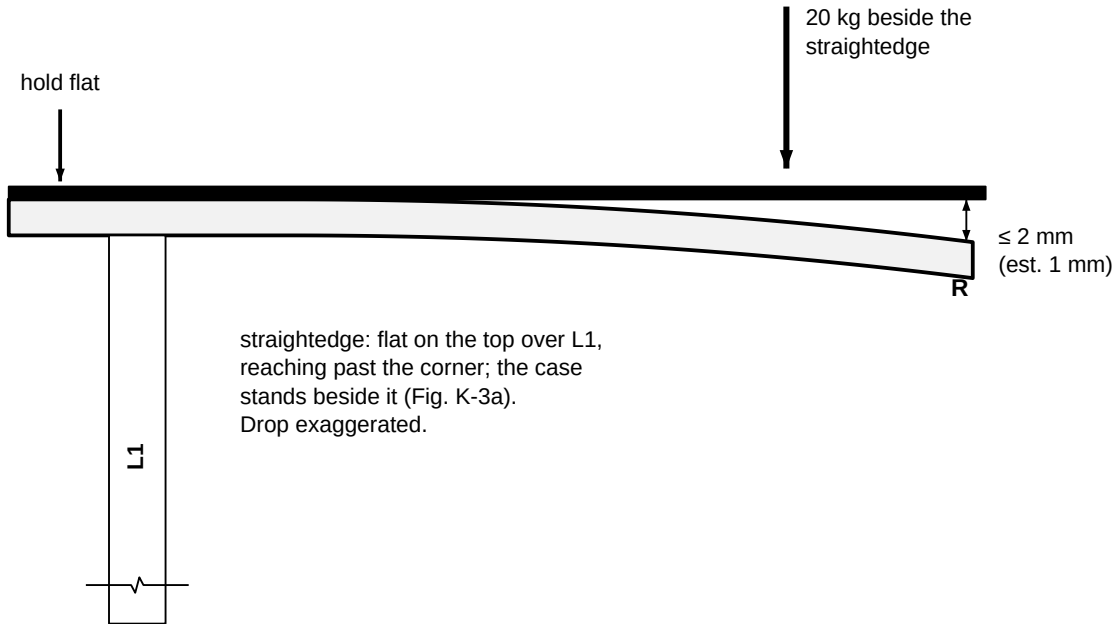


Fig. K-3b Elevation along the straightedge, from leg L1 to the corner. The case stands beside the straightedge, not under it.

Check 4 · All eight pads bear ACCEPTANCE CHECK

4

At each of the eight feet F1–F8, rock the pad sideways in its ring by hand. Do not turn it: turning changes its height. None rocks or lifts. A pad that rocks or lifts is not bearing: see chapter L, "Top moves on its ALEX".

Tests: the racking estimate (eight pads captive in their rings hold the frame against a sideways push) and the ALEX load share. STRUCTURAL ESTIMATE

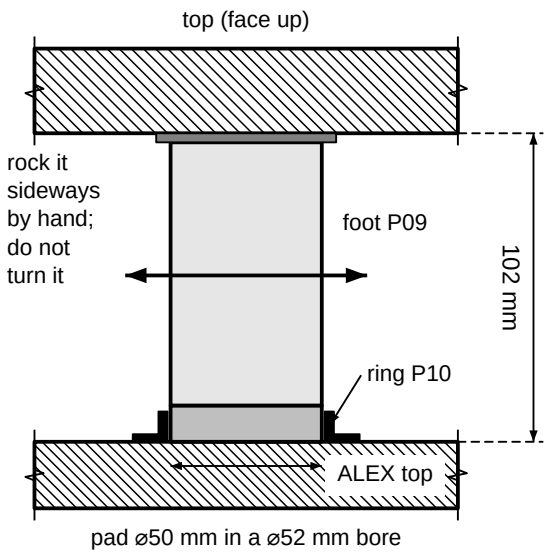


Fig. K-4 Foot P09 on its pad in ring P10 on the ALEX top.

Check 5 · Straightedge at the joint, now and later ACCEPTANCE CHECK

5

Lay a straightedge across the joint at the inside corner (Fig. K-2, position 4) and **write down any step**. Check it again after two weeks and after the first month. If the corner has settled, retighten the six fitting bolts (25 N·m, chapter L).

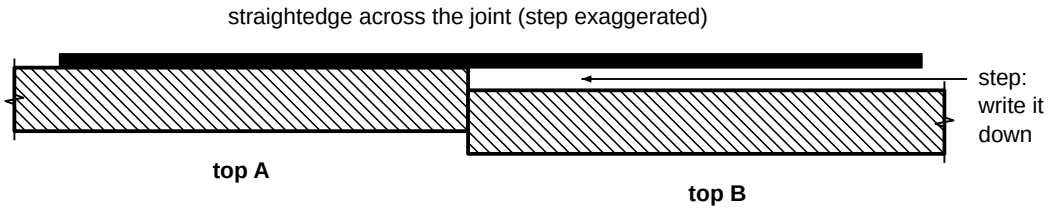


Fig. K-5 Section across the joint at the inside corner. Step drawn exaggerated.

Step at the joint	Day 0	Two weeks	One month	Bolts retightened?
mm, and which top is higher				

Tests: corner-joint slip, an estimate, not a measured capacity. STRUCTURAL ESTIMATE

Check 6 · Sideways push ACCEPTANCE CHECK

6 Push the desk sideways along each wall. No sway at the top beyond a few millimetres, and no top moves on its ALEX.

Tests: racking. The OLOV legs give almost no racking resistance; the two ALEX units, locked to the tops by the pads in their rings, do. STRUCTURAL ESTIMATE

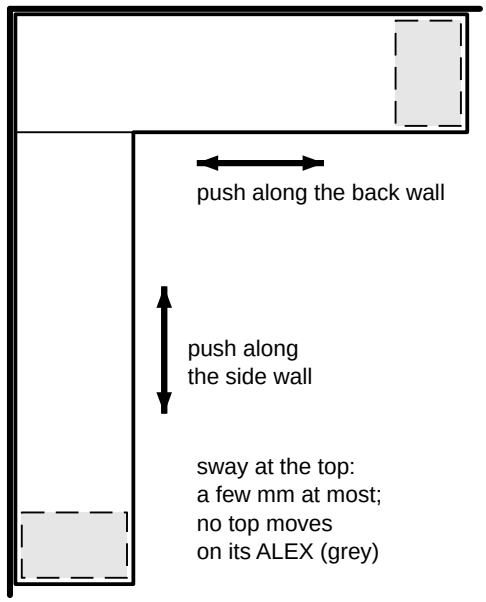


Fig. K-6 Push along the back wall, then along the side wall.

Check 7 · Screw tips ACCEPTANCE CHECK

7 **No screw tip** visible, or felt as a bump, on either laminate face. Run a hand over both tops, above every leg, foot plate, rail screw and joining plate.

Check 8 · Knees clear ACCEPTANCE CHECK

8 **Knees clear.** Sit at the corner and slide in: nothing within reach of a knee or thigh. The rails' lower edge is 75.1 cm from the floor, 12 cm back from the front edge.

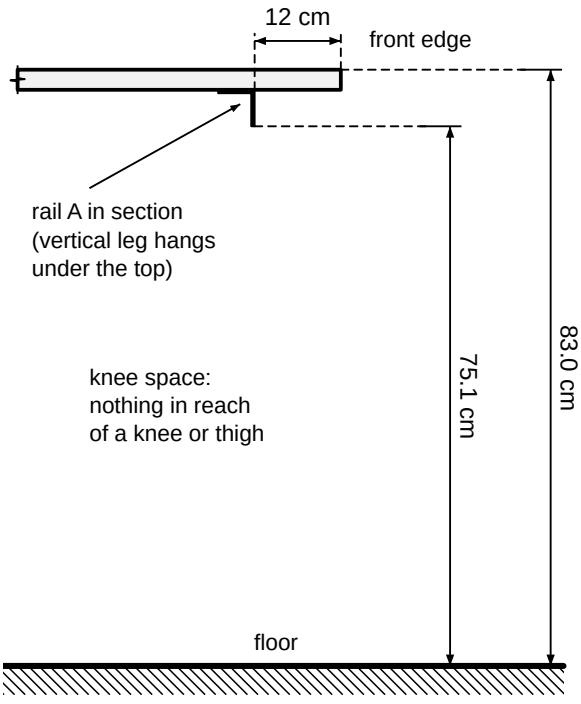


Fig. K-7 Section through top A's front at rail A.

Check 9 · ALEX fronts ACCEPTANCE CHECK

9

Both ALEX fronts open fully (ALEX 1's drawers, ALEX 2's door) without touching a foot or a leg.

Check 10 · Retighten ACCEPTANCE CHECK

10

Retighten every screw after two weeks, then repeat check 5.

Sign-off

#	Acceptance check	Pass when	Structural estimate it tests	Result
1	Surface height, six corners, then the inside corner	83.0 cm ± 0.3 cm each (85.0 / 87.0 / 89.0 cm with 2 / 4 / 6 cm ALEX feet)	—	
2	Level and fingertip across the joint	flat both ways; no step felt	—	
3	Corner load test, 20 kg	drop ≤ 2 mm; returns; no creak; nothing moved	sag at the inside corner (estimate about 1 mm); corner joint slip	
4	All eight pads bear	none rocks or lifts when rocked sideways (never turned)	racking; ALEX load share	
5	Straightedge at the joint	step written down; re-checked at two weeks and one month	corner joint slip (settling)	
6	Sideways push along each wall	sway a few mm at most; no top moves on its ALEX	racking	
7	Screw tips	none visible or felt on either laminate face	—	
8	Knees clear	nothing in reach; rail lower edge 75.1 cm from the floor, 12 cm back	—	
9	ALEX fronts	open fully, touch no foot or leg	—	
10	Retighten every screw	done after two weeks	—	

ACCEPTANCE CHECK pass or fail on this desk. STRUCTURAL ESTIMATE a number from the design review that the check confirms or refutes. If a check fails, see chapter L.

Checked by	Date	Two-week re-check	One-month re-check

L · Maintenance and troubleshooting

Schedule

When	Do	See
Day 0	Write down the step at the joint (straightedge at the inside corner).	Chapter K, check 5, Fig. K-5
Two weeks	Retighten every screw. Straightedge at the joint again; if the corner has settled, retighten the six fitting bolts to 25 N·m. Look at each printed ring and its three screws.	Chapter K, checks 5 and 10; troubleshooting
One month	Straightedge at the joint again; retighten the six fitting bolts if it settled.	Chapter K, check 5
Any time	If the corner ever drops a millimetre, retighten the six fitting bolts.	Corner bolts, below

> ! **NEVER DEEPER THAN 21 MM INTO A TOP**
> The tops are 28 mm particleboard with a thin laminate skin; a screw reaching the skin lifts it. Replace a screw only with the same size. If a top measures 27 mm, nothing changes: the deepest screw (rail #10 × 1 in) still stops 6 mm short of the laminate. Under 27 mm, stop and ask. SOURCE-DEFINED

Corner bolts

Bolts six 3/8-16 × 1-1/2 in hex bolts, grade 5 (H08), washer under head and nut (H09), nylon-insert nuts (H10): four through rail A's vertical leg, two through rail B's leg with a fitting on each side

Torque 25 N·m (18 ft·lb); without a torque wrench, pull as hard as one hand can on a 15 cm wrench, then stop

Why a friction joint: each fitting leg has two 21 × 12 mm slots, centred 29.15 mm and 78.45 mm from the heel VERIFIED ; they leave each bolt 2.5 mm of play across and 11.5 mm along, and the rails' 7/16 in holes 1.6 mm, so only the clamping force keeps rail B's corner from settling

Order tighten only with the two surfaces flush and rail B's end touching rail A

The mending plates across the joint align the tops; they do not carry the corner. SOURCE-DEFINED

Troubleshooting

Symptom	Cause	Fix
Corner has dropped; step at the joint grew	The six fitting bolts lost clamping force (friction joint); supports settling	Bring the surfaces flush with the OLOV feet and the round feet, never by packing anything under the joining plates. Check rail B's end still touches rail A. Retighten all six bolts to 25 N·m. Write the new step down (Fig. K-5).
Top moves on its ALEX, or a pad rocks or lifts	A pad is out of its ring or not bearing	Lift that ALEX end a few millimetres and slide the ALEX unit, never the top alone, until the pads drop into their rings (chapter G). Each foot set with the 102 mm foot height gauge, no rock. Level along x and y; re-check 83.0 cm at the corners (chapter K, Figs. K-1 and K-4).
Printed ring cracked or its screws loose	Ring wall overloaded by a sideways push	Inspect at two weeks. A crack: reprint the ring (parts/structure/foot-ring.stl), PETG or ABS, flange-down: 52 mm bore, 3 mm wall, 75 × 2.5 mm flange. Refit with #6 × 5/8 in flat-head screws (H06), 5/64 in pilots 12 mm deep (12 ALEX puck). Never run the desk with a ring missing.
Leg moves or wobbles	OLOV plate screws loose	Retighten the five M4 × 16 mm screws (H01). A screw that spins: repair the hole (below), re-pilot 1/8 in, 15 mm deep (15 LEG puck). Re-set the OLOV to 80.2 cm, or the ALEX-feet table's number (chapter F).
Monitor clamp creeps	Clamp loosened over time	Retighten the clamp with the crossbar fitted. Keep the steel plate (P11) flat on its outline under the jaw. Never add a soft pad under the jaw.
Screw spins (stripped)	Hole worn in the particleboard	Back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot.
A mark is off by more than 2 mm	Marking error	Redraw it. A stray 1/8 in hole in the underside does no harm.
A fitting bolt will not go in	Holes not quite in line	Open the hole in the rail with a round file; never force it.

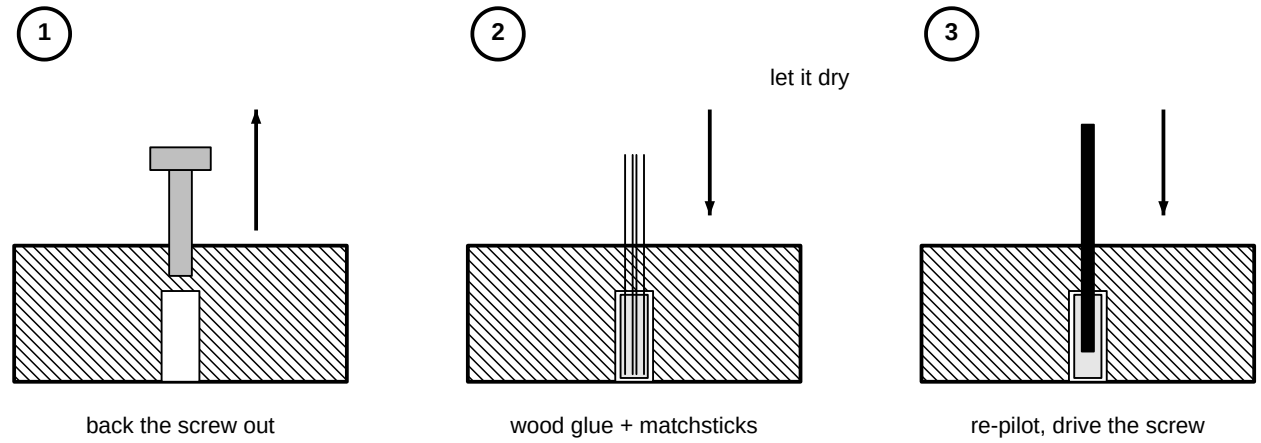


Fig. L-1 A screw that spins: back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot with the step's own bit and depth.

Every fix above is from the build manual and the build guide **SOURCE-DEFINED**.

Appendix · Outstanding measurements

These values were not measured on the purchased parts when this manual was built. Each is marked MEASURE ON HARDWARE at its point of use. Measure it before the step that uses it; if it differs, the step's own warning panel says what changes.

Measurement	Expected	Used in
ALEX bottom panel thickness (only for the optional ALEX feet)	not yet known	Chapter F, optional ALEX feet: three #6 × 5/8 in screws each. (The top panel is measured: 15.9 mm, VERIFIED.)
ALEX height, with and without its stock feet	70.0 cm (docs assume)	Chapter F, OLOV setting 80.2 cm; underside 80.2 cm; chapter K, check 1, surface 83.0 cm
Feet pad diameter (calipers)	50 mm; rings bored 52 mm	Chapter B, rings P10 before printing; chapter K, check 4
EKBACKEN thickness (calipers)	28 mm	Chapter D, every screw into a top (never deeper than 21 mm; at 27 mm nothing changes, under 27 mm stop and ask); chapter I, clamp opening 31 mm
VIVO top arm reach and maximum opening	reach about 89 mm; opening 31 mm or more	Chapter I: clamp zones, Fig. I-2
Delivered rail lengths and angle thickness	213 cm layout (bought as 84 in, 213.4 cm); 4.8 mm	Chapter E, hole stations; chapter F, rail pilot depth 25 mm from the steel face
Room: wall to wall, and baseboard thickness	252.0 cm and 317.0 cm	Chapter B, 252 × 317 cm of floor including the 3 cm wall gaps; chapter J, wall-gap trim
Feet load rating	unknown; the design wants 50 kg or more each	Load path through the eight feet ASSUMPTION / FIELD CHECK; chapter K, check 3 corner load test

Confidence labels

VERIFIED	measured on the purchased part, or taken from its own geometry file.
SOURCE-DEFINED	defined by the project documents (docs/build-manual.md). A source-defined dimension is never called verified.
ASSUMPTION / FIELD CHECK	engineering estimate or field check.
MEASURE ON HARDWARE	not yet measured: measure it before the step that uses it.

Quality control

Every number in this manual comes from manual/data.json (extracted from guide.html's D object), manual/facts.json or docs/build-manual.md. Source: HEAD of https://github.com/scotthowson/desk-setup on 23 Oct 2026 (supersedes a2e99ea with measured facts). If you change a measured value, change it in the docs first, then rebuild the manual.